How Neural Networks Work

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Introduction to Neural Networks

An artificial neural network is a computing system loosely inspired by the way neurons in the human brain are connected. It is the core technology behind deep learning, which powers image recognition, speech assistants, language translation and modern chatbots. A neural network learns by looking at many examples and slowly adjusting millions, or even billions, of numerical values called weights until its outputs become accurate. The human brain has about 86 billion biological neurons; artificial networks are far simpler, but their layered structure lets them learn very complex patterns.

History

The first mathematical model of a neuron was proposed in 1943 by Warren McCulloch and Walter Pitts. In 1958 Frank Rosenblatt built the Perceptron, a single-layer network that could learn to classify simple patterns. Interest faded after 1969, when Marvin Minsky and Seymour Papert showed the limits of single-layer perceptrons. In 1986 David Rumelhart, Geoffrey Hinton and Ronald Williams popularised the backpropagation algorithm, which made it possible to train multi-layer networks. A major breakthrough came in 2012 when AlexNet, a deep convolutional neural network, won the ImageNet competition. Geoffrey Hinton, Yoshua Bengio and Yann LeCun received the 2018 Turing Award for their work on deep learning, and Hinton shared the 2024 Nobel Prize in Physics with John Hopfield.

What Is a Neuron in a Neural Network?

A neuron, also called a node or unit, is the basic building block of a neural network. An artificial neuron receives one or more numerical inputs. Each input is multiplied by a weight, which expresses how important that input is. The neuron adds up all the weighted inputs and adds a number called the bias, which shifts the result. The sum is then passed through an activation function, and the output is sent to neurons in the next layer. In short, a neuron computes a weighted sum of its inputs plus a bias and then applies an activation function.

Layers of a Neural Network

Neurons are arranged in layers. The input layer receives the raw data, such as the pixel values of an image or the numbers describing a customer. One or more hidden layers sit in the middle and do the actual pattern learning; each hidden layer builds on the features found by the previous one. In an image network, early layers detect edges, middle layers detect shapes and later layers detect objects such as faces. The output layer produces the final answer, such as the probability that the image shows a cat. A network with many hidden layers is called a deep neural network, which is why the field is called deep learning.

What Does an Activation Function Do?

An activation function decides how strongly a neuron should "fire" and, most importantly, introduces non-linearity into the network. Without activation functions, a network with many layers would behave like a single linear equation and could not learn curved or complex relationships. Common activation functions are ReLU (Rectified Linear Unit), which outputs zero for negative inputs and the input itself for positive inputs and is the most popular choice in hidden layers; the sigmoid function, which squeezes values between 0 and 1 and is useful for yes-or-no probabilities; tanh, which outputs values between -1 and 1; and softmax, which turns a list of numbers into probabilities that add up to one and is used in the output layer for multi-class classification.

How a Network Learns: Forward Pass, Loss and Backpropagation

Training happens in repeated cycles. In the forward pass, input data flows through the layers and the network produces a prediction. A loss function then measures how far the prediction is from the correct answer; for example, mean squared error is used for regression and cross-entropy for classification. Next comes backpropagation, short for "backward propagation of errors". Backpropagation is the algorithm that calculates how much each weight contributed to the error by moving backward from the output layer to the input layer, using the chain rule of calculus to compute gradients. An optimiser such as gradient descent, SGD or Adam then updates each weight a small step in the direction that reduces the loss. The size of that step is controlled by the learning rate. One full pass through the entire training dataset is called an epoch, and training usually runs for many epochs until the loss stops improving.

Main Types of Neural Networks

A feed-forward network passes information in one direction and suits tabular data. A Convolutional Neural Network (CNN) uses filters that slide over an image and is the standard choice for computer vision. A Recurrent Neural Network (RNN), and its improved forms LSTM and GRU, handle sequences such as text and time series by keeping a memory of earlier inputs. The Transformer, introduced by Google researchers in the 2017 paper "Attention Is All You Need", uses a mechanism called self-attention and is the foundation of modern large language models. Generative Adversarial Networks (GANs) and diffusion models generate new images.

Uses and Indian Examples

Neural networks recognise faces for phone unlock, read handwriting, transcribe speech, translate languages, detect cancer in scans and drive autonomous cars. In India, neural networks power Aadhaar-related face authentication, speech recognition for Indian languages in Bhashini, number-plate recognition on highways, and models that detect crop diseases from leaf photos. Researchers at institutes such as IIT Madras and Anna University develop neural networks for small-object detection and medical imaging.

Challenges and Future

Neural networks need large datasets and powerful hardware, can overfit, and often act as "black boxes" whose decisions are hard to explain. Techniques such as dropout and regularisation reduce overfitting, while explainable AI tries to reveal why a network gave an answer. The future points toward smaller, efficient networks that run on phones and edge devices, and toward brain-inspired neuromorphic chips.

Quick Facts

- A neuron computes a weighted sum of its inputs plus a bias and then applies an activation function.
- An activation function adds non-linearity so networks can learn complex patterns; ReLU is the most popular.
- Backpropagation calculates the error contribution of each weight by moving backward through the network.
- The Perceptron was built by Frank Rosenblatt in 1958.
- The Transformer architecture was introduced in 2017 in the paper "Attention Is All You Need".